torch.nn.utils.rnn.pad_sequence#

https://pytorch.org/docs/stable/generated/torch.nn.utils.rnn.pad_sequence.ht

Description:

Pad a list of variable length Tensors with padding_value. pad_sequence stacks a list of Tensors along a new dimension, and pads them to equal length. sequences can be list of sequences with size L x *, where L is length of the sequence and * is any number of dimensions (including 0). If batch_first is False, the output is of size T x B x *, and B x T x * otherwise, where B is the batch size (the number of elements in sequences), T is the length of the longest sequence. This function returns a Tensor of size T x B x * or B x T x * where T is the length of the longest sequence. This function assumes trailing dimensions and type of all the Tensors in sequences are same.

Function Signature

```
torch.nn.utils.rnn.pad_sequence(sequences, batch_first=False, padding_value=0.0, padding_side='right')
[source]#
Parameters
Returns
Return type
```

Returns:

Tensor

Code Examples

```
>>> from torch.nn.utils.rnn import pad_sequence
>>> a = torch.ones(25, 300)
>>> b = torch.ones(22, 300)
>>> c = torch.ones(15, 300)
>>> pad_sequence([a, b, c]).size()
torch.Size([25, 3, 300])
```

Notes & Warnings

NOTE

Note This function returns a Tensor of size T x B x * or B x T x * where T is the length of the longest sequence. This function assumes trailing dimensions and type of all the Tensors in sequences are same.

Detailed Documentation

Documentation

Pad a list of variable length Tensors with padding_value.

pad_sequence stacks a list of Tensors along a new dimension, and pads them to equal length. sequences can be list of sequences with size L x *, where L is length of the sequence and * is any number of dimensions (including 0). If batch_first is False, the output is of size T x B x *, and B x T x * otherwise, where B is the batch size (the number of elements in sequences), T is the length of the longest sequence.

This function returns a Tensor of size T x B x * or B x T x * where T is the length of the longest sequence. This function assumes trailing dimensions and type of all the Tensors in sequences are same.

sequences (list[Tensor]) – list of variable length sequences.

batch_first (bool, optional) – if True, the output will be in B x T x * format, T x B x * otherwise.

padding_value (float, optional) – value for padded elements. Default: 0.

padding_side (str, optional) – the side to pad the sequences on. Default: 'right'.

Tensor of size T x B x * if batch_first is False. Tensor of size B x T x * otherwise